

Rishi Rajendra Gandhi

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Research Interests

Bioprospecting extremophilic enzymes for industrial lignocellulose degradation; microbial adaptation mechanisms in poly-extreme environments; astrobiology and biosensor development.

Education

- **CSIR - National Institute of Oceanography, Goa,**
Degree - AcSIR Ph.D. **Aug. 2021 - Present**
- **Indian Institute of Science Education and Research Bhopal,**
Degree - BS-MS (Integrated Bachelor of Science and Master of Science),
Major - Biological Sciences. **Aug. 2015 - May. 2020**

Fellowships & Achievements

- **Graduate Aptitude Test in Engineering (GATE)** **Feb 2020**
Secured AIR 170.
- **CSIR NET Junior Research Fellowship** **Jun 2019**
Secured AIR 147.
- **INSPIRE-Scholarship for Higher Education (SHE)** **Aug 2015 - Jun 2020**

Projects & Internships

- **AcSIR Ph.D. Scholar** **Duration:** Aug 2021 - Present
Dr. Rakhee D Sinai Khandeparker, Senior Principal Scientist,
CSIR - National Institute of Oceanography.
Title: **Acido-tolerant thermophilic lignocellulolytic enzymes from marine microbes**
Summary: Characterisation of poly-extremophilic cellulases and laccases (structural & functional profiling). Optimisation of enzyme efficacy for efficient lignocellulose saccharification for biofuel production.
- **MS thesis** **Duration:** May 2019 - Jun 2020
Dr. Varun Chaudhary, Assistant Professor,
Indian Institute of Science Education and Research, Bhopal.
Title: **Developing a Split-GFP complementation based sensor for canonical Wnt signaling activity in *Drosophila* S2R+ cells.**
Summary: Engineered a split-GFP biosensor to generate the first live fluorescent readout of canonical Wnt pathway activation in *Drosophila*.
- **Winter Internship** **Duration:** Nov 2018 - Dec 2018
Dr. Nagarjun Vijay, Assistant Professor,
Indian Institute of Science Education and Research, Bhopal.
Title: **Isolation and Characterization of novel tardigrades around the central region of India.**
Summary: Surveyed ponds, lichens, and mosses across Bhopal for undescribed tardigrade species.
- **iGEM Competition 2018** **Duration:** Feb 2018 - Nov 2018
Principal Investigators: Dr. Vineet Sharma, Dr. Chandan Sahi and Dr. Lavanya Bhagvatula,
Indian Institute of Science Education and Research, Bhopal.
Title: **Methnote: A prototype to methane sensor.**
Summary: Designed and built **MethNote**, a field-deployable engineered *E. coli* biosensor for detecting local methane leaks. Led scientific illustration and visual communication for the project. Secured **Bronze Medal** at the Giant Jamboree, Boston.

Research Papers & Reviews

- Palekar, K., Khandeparker, R.D.S., **Gandhi, R.**, Mahale, A. "From Waste to Wealth: Optimizing Fish Silage for Aquafeed and Environmental Impact" Waste and Biomass Valorization (2026). <https://doi.org/10.1007/s12649-026-03551-w>
- Wasnik, H., **Gandhi, R.**, Patil, N., Behera, R., Golait, A., Patel, T. and Baruah, T., 2021. A Comprehensive Review of Molecular Biology and Genetics of Cataract. International Journal for Research in Applied Sciences and Biotechnology, 8(2), pp.11-20. <https://doi.org/10.31033/ijrasb.8.2.2>

Contributor (Multi-Institutional Studies)

- **Early career ocean professionals' declaration contributor in**
Li, S., Addey, C. I., Roman, R., Hayashida.... **Gandhi, R.** "Early career ocean professionals' declaration on ocean negative carbon emissions for our ocean and future." The Innovation 6.9 (2025). <https://doi.org/10.1016/j.xinn.2025.101007>
- **iGEM Interlab Study contributor in**
Beal, J., Farny, N.G., Haddock-Angelli, T., Selvarajah, V., Baldwin, G.S., Buckley-Taylor, R., Gershater, M., Kiga, D., Marken, J., Sanchania, V. and Sison, A., 2020. Robust estimation of bacterial cell count from optical density. Communications Biology, 3(1), pp.1-29. <https://doi.org/10.1038/s42003-020-01127-5>.
- **iGEM Interlab Study contributor in**
Beal J, Baldwin GS, Farny NG, Gershater M, Haddock-Angelli T, Buckley-Taylor R, et al. (2021) Comparative analysis of three studies measuring fluorescence from engineered bacterial genetic constructs. PLoS ONE 16(6): e0252263. <https://doi.org/10.1371/journal.pone.0252263>.

Under Review

- **Gandhi, R.**, Khandeparker, R.D.S., Nikhita P. R., Paingankar, M., "Mangrove Ecosystem Degradation Restructures Bacterial Community Composition and Metabolic Function: A Metagenomic Analysis", Estuarine, Coastal and Shelf Science (Under Review Since Dec 2025).
- **Gandhi, R.**, Khandeparker, R.D.S., Nikhita P. R., Chudasama, K., Chudasama "Mangrove Health Shapes Lignocellulolytic Bacterial Communities", Letters in Applied Microbiology (Under Review Since Feb 2026).

Book Chapter

- Wasnik H., **Gandhi, R.**, Lucinda E., Gupta I. (2020). Soil microbiome of Cold Deserts and their biotechnologically useful adaptations for a life on Earth and beyond with Atacama desert as a special case study. (Yet to come in online domain) <https://doi.org/10.1201/9781003354031-4>

Presentations & Conferences

- **Oral presentation:** Feb 2026
World Ocean Science Congress 2026, Goa, India.
Presented **"Forged in Fire and Acid: Machine Learning Blueprints for Bacterial Poly-Extremozyme"**
- **Oral and Poster presentation:** Sept 2023
Samudramanathan 2023, Panaji, India.
Presented **"Poly-extremophiles: Exploring their diversity and potential for industrial applications."**
- **Oral presentation:** Sept 2022
World Biotechnology Industry Conference-2022, Qingdao, China.
Presented **"Mysteries of Poly-Extremophiles for sustainable future"**
- **EO Dashboard Hackathon** 23 Jun 2021 - 29 Jun 2021
Project Elevate competed amongst 233 projects and was one of 32 Global finalists. Explored COVID-19 impacts on tribal communities via EO satellite data (NASA, ESA, JAXA).
- **Oral and Poster presentation:** Oct 2018
iGEM Giant Jamboree 2018, Boston, USA.
Presented **"MethNote - A prototype of Methane Biosensor"** and secured Bronze Medal.

Workshops	• Skill India, 10 - 14 Feb 2025 CSIR-NIO, Goa, India Hands-on workshop on "Horizontal Gene Transfer (Protoplast Fusion)". • Skill India, 29 - 31 Jan 2024 CSIR-NIO, Goa, India Hands-on workshop on "Mass Spectrometry for Proteomics".
Courses	• Analog Astronaut Oct 2021 Participated under the guidance of Dr. Julio Rezende, in a staged Mars analogue mission focused on sustainability strategies for future space habitats, including habitat design R&D and facility upgrades at Habitat Marte, Brazil. • Introduction to Astrobiology Sept 2020 - Oct 2020 Amity Centre of Excellence in Astrobiology, Amity University Mumbai, India. • Astrobiology Australasia Meeting (AAM) Sept 2020 Online.
Technical Skills	• Microbiology: Extremophile isolation, enrichment culture, enzyme assays, protoplast fusion, molecular cloning • Molecular Biology: Split-GFP complementation, gene cloning, cDNA synthesis • Biochemistry: Enzyme kinetics, SDS-PAGE • Bioinformatics: Sequence analysis, phylogenetic analysis • Computational: Python and R for statistical analysis; machine learning (XGBoost) for enzyme activity prediction; response surface methodology for process optimization • Analytical: Mass spectrometry (proteomics), microscopy (light, fluorescence)
Leadership & Community Services	• NoRCEL SESSIONS Manager Feb 2026 - Present - Conducting Competitions for the NoRCEL • General Secretary, Sethu (CSIR-NIO Goa) Jul 2022 - Jul 2023 As Student Representative, launched: – Samudra Manthan'23 (First National Oceanography Scholars' Meet) – Health outreach: Breast camp, Senior Citizen Camp, One week-one lab, Ocean Day. • Biology Club Coordinator (IISER Bhopal) Aug 2017 - Aug 2018 Initiated: – "Tag Me If You Can" tree-awareness drive – Synthetic biology session series (later led to iGEM team).
Professional Memberships	• Ocean Society of India Dec 2022 - Present • Mangrove Society of India Jul 2022 - Present • Astrobiology Society of Britain Jun 2022 - Jun 2023 • Peer Reviewer, Science Publishing Group Nov 2021 - Present